Paramveer Singh

Full-Stack, Cloud & Distributed Systems Engineer | AI & Open-Source Developer

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LinkedIn • GitHub (Paramveersingh-S) • Portfolio • LeetCode • Substack

Summary

Full-Stack, Cloud, and Distributed Systems Engineer specializing in production **Go, Python, TypeScript/React/Next.js**, and **AWS/Terraform** infrastructure. Open-Source Developer in the **vLLM Data Plane/Networking Working Group**, AI Researcher at the **University of Tokyo's Matsuo-Iwasawa Lab (Ranked #1 globally among 2,000 participants)**, and author of **55+ merged upstream pull requests** across **vLLM, Unikraft, CERN, CNCF Meshery, Google DeepMind, and Scikit-HEP**.

Experience

vLLM Project (Data Plane / Networking Working Group) **Oct 2026 – Present**
Open Source Developer (Go, Python, Valkey/Redis, Kubernetes) *Remote*

- Reduced streaming Time-To-First-Token (TTFT) latency by decoding Chat opening deltas immediately without holding buffers (PR #4911), and maintained streaming protocol parity for DeepSeek V3/R1 reasoning content (`PR #3553`) and Mistral logprob chunks (PR #4818).
- Resolved distributed race conditions in Valkey cache access tracking via atomic operations without resurrecting expired keys (PR #4891) and enforced authorized `max-age` access controls on semantic cache reads (PR #4848).
- Preserved Anthropic cache-write tiers through settlement and pricing (PR #4939), eliminated zombie routes during runtime recovery (PR #4867), validated default decision deployments at config load to prevent panics (`PR #5294`), added Windows CLI locking guards (PR #5093), and shrank Kubernetes container image footprints in AIBrix (PR #2863).

Matsuo-Iwasawa Lab, The University of Tokyo (U-Tokyo) **Sep 2026 – Present**
Artificial Intelligence Researcher Intern *Tokyo, Japan (Remote)*

- **Ranked #1 globally among 2,000 participants** in the internal engineering competition; graduated with **Extraordinary and Honours** distinction.

Layer5 (CNCF Meshery) & FOSSASIA **Sep 2026 – Oct 2026**
Open Source Contributor (Go, Python, Cloud-Native Infrastructure) *Remote*

- Contributed to **CNCF Meshery** (`mesheryctl` CLI in Go, server preference store safety, and operator CRD documentation, PR #22300) and **FOSSASIA** open-source cloud and web initiatives; also merged **8 Go PRs** into **Unikraft's** `kraftkit` unikernel CLI (PR #2946–#2953).

CERN (European Organization for Nuclear Research) **Jul 2026 – Oct 2026**
Open Source Developer — `OpenPHIGS` & `TIGRE` (C/C++, Python, OpenGL/EGL, CI) *APAC (Remote)*

- **Modern Export & Headless Rendering:** Built a glTF 3D scene-export workstation and headless EGL backend for CERN's `OpenPHIGS` 3D detector display library (DELPHI, OPAL), enabling SVG/PDF/glTF generation on headless CI servers without X11/Wayland (PR #54).
- **Rendering Core:** Replaced legacy `glBegin(GL_POLYGON)` calls with a GLU tessellation pipeline fixing concave polygon crashes (PR #31, #36), and resolved severe parallel-beam stripe artifacts in `TIGRE` (PR #752, #748, #749).
- **Build, QA & Docs:** Integrated AddressSanitizer CI to catch memory leaks/use-after-free bugs, streamlined the 4-platform CI matrix (PR #53), refactored C/Fortran bindings (PR #51), and automated CMake Doxygen UNIX man page generation (PR #32).

Bharat Electronics Limited (BEL) **Jun 2026 – Aug 2026**
Research and Development Intern — Testing Department *Panchkula, India*

- Executed RF power and Automated Test Procedures (ATP) on defense/railway UHF Data Radios (DPWCS CULM 207, STARS-V Mk II), verifying ±1.5 ppm frequency accuracy, 0% Packet Error Rate across 90 dB attenuation rigs, SINAD receiver sensitivity down to −115 dBm, and MIL-STD-810G compliance.

GirlScript Summer of Code (GSSoC) **May 2026 – Aug 2026**
Open Source Developer — AI Track *Hybrid*

- Selected AI/ML open-source contributor; optimized model inference pipelines and conducted code reviews with a global developer community.

CSIR – Central Scientific Instruments Organisation (CSIO) **Jun 2025 – Jul 2025**
Summer Software Development Intern (React.js, Next.js, MySQL) *Chandigarh, India*

- Engineered and deployed the full-stack ISTC-CSIR Government Seat Allocation Web Portal (**React.js, Next.js, MySQL**) featuring candidate/admin portals, **JWT authentication**, automated seat-allocation algorithms, and real-time analytics dashboards.

Yamada Automation **Aug 2022 – Sep 2022**
Internship Trainee *India (On-site)*

- Trained in Japanese industrial automation workflows, precision CNC programming, and CAD/CAM manufacturing systems.

Additional Open-Source Contributions

- **Google DeepMind (`torax`, `formal-conjectures`) & Google Research (`tabfm`)** [#2242] [#4416] [#4418] [#50]
  Patched initial-condition plotting and empty dataset handling in DeepMind's `torax` fusion simulator (`#2242`), aligned PyTorch TabFM checkpoint configurations (`#50`), and formalized computational complexity statements in Lean 4 (`#4416`, `#4418`).
- **Scikit-HEP (`uproot5`, `awkward`), CambridgeCIA (`LION`) & FOSS United** [#1667] [#4150] [#188] [#1631]
  Deduplicated redundant **Vue.js** session profile API requests on the FOSS United production dashboard (`#1631`), engineered backward-compatible C++ patches in `uproot5` (`#1667`), added type annotations in `awkward` (`#4150`), and hardened geometry initializers in CambridgeCIA's `LION` (`#188`, `#190`).

Key Engineering Projects

GateKeeper (Cloud-Native LLM API Gateway) — *Go, Redis, NATS JetStream, PostgreSQL, Next.js 15, Terraform, AWS EKS* [GitHub]

- Built a high-concurrency API Gateway and intelligent rate limiter in **Go** with token-aware rate limiting, multi-provider failover routing, asynchronous cost settlement via NATS JetStream, and a **Next.js 15** control plane.
- Sustained $>$ 5,000 **RPS** with $<$ 10 **ms p50 latency** overhead; implemented Semantic Vector Caching (Redis `FT.SEARCH`) yielding $<$ 5 **ms responses** and 100% token cost savings on repeated queries, instrumented with OpenTelemetry and Jaeger.

Serverless Pastebin (Zero-Trust AWS Cloud Infrastructure) — *Terraform (HCL), AWS Lambda, Python 3.12, DynamoDB, CloudFront* [GitHub]

- Architected a 100% modular **Terraform**-provisioned serverless platform on **AWS**: Python 3.12 Lambda functions behind API Gateway HTTP APIs, DynamoDB with native TTL expiration, and an S3 + CloudFront CDN frontend secured via Origin Access Control (OAC).
- Enforced keyless GitHub Actions CI/CD via **OIDC federation**, least-privilege IAM policies, CloudTrail auditing, and CloudWatch P95 latency alarms.

Receipt (Autonomous Agent PR Verification & Memory Platform) — *Python 3.12, FastAPI, FastMCP, Next.js 16 (TypeScript), OpenAPI/SSE* [GitHub]

- Engineered a full-stack PR verification service and Next.js 16 dashboard streaming real-time telemetry over Server-Sent Events (SSE) and exposing institutional memory via Model Context Protocol (MCP), backed by 65+ pytest suites (**88% coverage**).

Semantic-Search-RAG (Distributed Enterprise Search) — *Python, FastAPI, Kafka, PySpark, Elasticsearch, pgvector, gRPC, Docker* [GitHub]

- Built a distributed hybrid search microservices platform (BM25 + Dense Vectors + custom SPLADE v2) over an Apache Kafka + PySpark streaming pipeline, delivering **sub-100 ms** answer generation via decoupled gRPC embedding services.

FlashAttention-2, Triton Fused Kernels & Custom HGEMM — *CUDA C++, OpenAI Triton, PyTorch, Tensor Cores* [Flash-Atten2] | [Triton] | [HGEMM]

- Engineered FlashAttention-2 and a 7-phase HGEMM kernel from scratch in CUDA C++ plus fused Triton SRAM kernels, achieving **14.0x speedup** on BERT-large, **13.3x speedup** on LLaMA-7B prefill, **96.3% TTFT reduction**, and **13.3 TFLOPS (70.2% cuBLAS efficiency)** on NVIDIA T4.

Publications & Scholastic Achievements

- **IEEE-ICSPED 2025 (Cryptography & Security)**: *"CHRONOS: A Chaos-Harnessed TRNG for Neural Initialization and Secure Key Synthesis"* — Designed a True Random Number Generator passing all **15 NIST SP 800-22** cryptographic randomness suites.
- **Springer Journal 2025**: *"AI-Driven Sanskrit Character Recognition and Translation to Text and Audio Formats"* — End-to-end OCR, custom HuggingFace seq2seq, and Vedic TTS pipeline.
- **Awards**: **Ranked #1 / 2,000 Globally** (Matsuo-Iwasawa Lab, U-Tokyo, 2026) • **NASA Space Apps Challenge 2025 Global Nominee & Regional Winner** (top among 18,868+ teams) • **1st Place Overall** at IPD EXPO 2025 • **National Finalist** at India Innovates 2026 (35,000+ entries).

Skills, Education & Certifications

Languages: Go (Golang), Python, TypeScript, JavaScript (ES6+), C/C++, CUDA C++, SQL, HCL (Terraform), Bash/Shell

Backend, Frontend & Cloud: FastAPI, Go (`net/http`, Gin), React.js, Next.js, Vue.js, AWS (Lambda, API Gateway, EKS, S3, DynamoDB, CloudFront, IAM/OIDC, CloudWatch), Terraform, Docker, Kubernetes, PostgreSQL, MySQL, Redis/Valkey, Kafka, NATS JetStream, gRPC

Education: B.E. in Electronics & Communication Engineering, CCET, Panjab University (Aug 2024 – 2027 Exp., CGPA: 7.4/10) • **Diploma in Mechanical Engineering (Tool & Die)**, CSIR-CSIO Indo-Swiss Training Centre (2020 – 2023, CGPA: 8.5/10)

Certifications: Linux Kernel Development (**LFD103**, Linux Foundation) • Quantum Cryptography (**QNu Labs**) • Model Context Protocol (**Hugging Face**)